

Gowri Priyanka Gottapu

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AL/ML Engineer | Gen AI Engineer | Data Engineer | Python Developer

AI/ML & Data Engineer with 6.8 years of experience designing scalable GenAI and ML solutions. Improved model precision by 35% and reduced ETL latency by 45% within 2 quarters. Skilled in Python, Spark, RAG pipelines, LLM integration, and API design. Delivered real-world, secure AI deployments across cloud-native and enterprise platforms.

WORK EXPERIENCE

enGen

05/2024 - Present

AI/ ML Engineer / Python Developer

Pittsburgh, Pennsylvania

- Led the design and deployment of an LLM-powered internal assistant for claims triage, reducing processing time by 45% in 4 months and boosting compliance accuracy by 30%.
- Developed scalable RAG pipelines using FAISS and LangChain to retrieve and ground patient interaction data, improving GenAI model accuracy by 38%
- Built and deployed Python-based ML pipelines that reduced data preprocessing time by 40% and improved pipeline stability across distributed environments.
- Integrated LLMs with identity platforms (Okta, Azure AD) to automate user provisioning and threat alert summarization, reducing manual SOC workload by 35%.
- Orchestrated Airflow DAGs for daily model scoring and ETL tasks, cutting failed jobs by 50% within two sprints and improving job SLA compliance.
- Participated in architecture design sessions for multi-cloud ML systems, aligning infrastructure decisions across AWS and Azure.
- Conducted regular knowledge-sharing sessions with analysts and business teams to align AI initiatives with compliance and operational goals.
- Implemented vector store architecture for real-time document embedding and retrieval, cutting document lookup time by 60% in internal systems.
- Fine-tuned open-source LLMs using domain-specific datasets, increasing accuracy on classification and summarization tasks by 25–30%.
- Collaborated with cross-functional DevSecOps and MLOps teams to deliver secure ML deployments on AWS and Azure in under 5 weeks, meeting internal audit standards.
- Automated model monitoring workflows using Prometheus and custom Python validators, reducing anomaly detection delay by 40%.
- Integrated Generative AI modules for summarizing patient records, reducing manual review time by 40% within 3 months.

Charter Communications

02/2023 - 04/2024

Generative AI Engineer/ AIML Engineer

Charlotte, NC

- Designed and managed databases, creating Python-based RESTful APIs using Flask, SQLAlchemy, and PostgreSQL.
- Contributed to the full Software Development Life Cycle (SDLC) for the application's development.
- Installed and configured Apache Airflow for S3 bucket and Snowflake data warehouse and created dags to run the Airflow.
- Led the design and maintenance of large-scale data processing pipelines with PySpark and Dask, ensuring efficient data flow and integration across analytics platforms within 6 months.
- Developed prompt-engineered workflows to generate personalized content recommendations and automate knowledge base responses using telecom domain data.
- Engineered custom feature stores for ML models that predict viewer preferences, network anomalies, and subscriber segmentation based on usage behavior.
- Deployed and monitored AI/LLM services on cloud-native infrastructure (AWS/GCP), with auto-scaling, logging, and alerting enabled through MLflow and Prometheus.

- Engineered data pipelines with PySpark and Airflow, boosting processing efficiency by 25% within 6 months for AI/ML applications.
- Built and deployed LLM-based chatbots and assistants for customer service automation, reducing average handling time (AHT) by 35% and increasing CSAT scores.
- Used generative AI to create synthetic customer personas and simulate user journeys for A/B testing and audience modeling.
- Reduced customer service AHT by 35% within 2 quarters by deploying LLM-based chatbots and assistants.
- Participated in the development of the application's front-end using Python 2.7, HTML5, CSS3, AJAX, JSON, and jQuery.
- Developed and managed analytical solutions by building PowerPoint presentations and creating visualization results with tools like Tableau and Power BI.
- Created Bitbucket repositories and Managed Bitbucket user access and branch permissions.
- Developed Web Services using SOAP for data exchange in XML format, employing Test-Driven Development (TDD) for required services.
- Developed MapReduce/Spark Python modules for machine learning & predictive analytics in Hadoop on AWS
- Worked on AWS Redshift and RDS for implementing models and data on RDS and Redshift.
- Led design and maintenance of large-scale data pipelines using PySpark and Dask, streamlining processing of 10TB+ data monthly across analytics and visualization platforms.

CVS

09/2021 - 08/2022

Sr. Python Developer

Woonsocket, RI

- Implemented business logic in Python to prevent, detect, and manage duplicate member entries.
- Automated resulting scripts and workflow using Apache Airflow and shell scripting to ensure daily execution in production.
- Worked on data processing and transformations and actions in spark by using Python (Pyspark) language.
- Applied Python core concepts, improving code efficiency and performing statistical analyses using libraries within 3 months.
- Developed Python scripts to use AWS CLI for consistent tagging across resources, improving resource organization and tracking.
- Migrated applications to AWS, optimizing S3 bucket policies for storage and backup, improving data security and access within 3 months.
- Built the entire frontend and backend modules using Python on the Django Web Framework.
- Responsible for maintaining the code base for each project using Version Control tools GitHub and GitLab
- Worked with JSON-based REST Web Services and Amazon Web Services (AWS).
- Automated ETL workflows using Python and Apache Airflow, reducing manual intervention and increasing data refresh reliability by 40%.
- Wrote Python scripts to parse XML documents and load data into the database.
- Developed a Python-based RESTful API for sales tracking and analysis using Flask, SQL Alchemy, and PostgreSQL.
- Developed and maintained RESTful APIs using Flask and FastAPI, improving application response time by 30% within 3 months.

Jefferies Group

05/2020 - 08/2021

Python Developer/ Data Engineer

NewYork

- Created automated data validation and cleansing scripts using Python and libraries such as Great Expectations to maintain high data accuracy and consistency for analytics.
- Developed a robust database management strategy that improved data retrieval efficiency and security, fostering seamless integration across applications.
- Architected and optimized ETL workflows in the AWS environment using Python, Boto3, and PySpark,
- Integrated Apache Airflow with AWS to monitor multi-stage ML workflows with the tasks running on Amazon SageMaker.
- Created data pipelines for different events to load the data from DynamoDB to AWS S3 bucket and then into HDFS location

- Deployed AWS CloudFormation templates in Python and utilized Python SDKs with AWS Organizations to ensure consistent infrastructure practices and bolster centralized security.
- Developed RESTful APIs using Golang, integrating them with various application landscapes.
- Facilitated cross-team collaboration to align backend services with frontend needs, resulting in a cohesive user experience and increased operational efficiency.

Jobiak

12/2018 - 04/2020

Python Developer/ Data Analyst

Hyderabad, India

- Developed Python scripts for efficient data extraction from CSV, XML, and JSON, streamlining database loading and REST API integration within 3 months.
- Streamlined daily tasks and CI/CD pipelines using Python, enhancing update efficiency within 3 months, while generating critical reports with NumPy and Matplotlib.
- Involved in writing unit testing code using Unit Test/PyTest and integrating the test code with the build process additionally involved in working with Python ORM Libraries including Django ORM, SQLAlchemy
- Collaborated with cross-functional teams to integrate REST-based microservices, streamlining data access and improving overall application performance.
- Created ETL programs for data transition from DynamoDB to AWS Lambda and Snowflake, streamlining processes with Python functions.
- Developed a fully automated continuous integration system using Gerrit, Jenkins, AWS Glue MySQL and GIT, Amplify custom tools developed in Python

EDUCATION

Masters in Computer Information Systems

Rivier University

NH, USA • 01/2024

Bachelors in Computer Science Engineering

Vignan's Institute of Engineering for Women

Vishakhapatnam, India • 01/2018

SKILLS

Programming Languages: C, C++, Core Java, Golang, Java, JavaScript, PHP, Python, R, SQL Shell programming

Frameworks: CSS Bootstrap, Django, Docker, Flask, Flask Web framework, Kubernetes, Rest Framework, SAS

Databases: MongoDB, MySQL, NoSQL DBs like Dynamo DB, Oracle, SQL databases like MS-SQL, SQL Server

Cloud Technologies: AWS, Azure

Bug Tracking Tools: Bugzilla, JIRA, MATLAB

Data Streaming & Messaging: Kafka, RabbitMQ

Scripting Languages: AJAX, CSS, JavaScript, jQuery, PHP, PowerShell Scripting

Version Control: Amplify & GIT, CVS, GitHub, GitLab, SVN

Operating Systems: Deployment Tools: Ansible, Heroku, Jenkins

Operating Systems: Linux, Mac OS, UNIX, Windows

Methodologies: Agile, Iterative, Scrum, Waterfall

API Integration: JSON, REST, SOAP, XML

ORM: SQLAlchemy

IDE/Development Tools: Eclipse, PyCharm, VSC

Other Tools: Excel, MS Word, Oracle data Management suite, PowerPoint, Salesforce CRM, SharePoint, Swim charts, Visio